TED UNIVERSITY

Department of Computer Engineering

CMPE 491

Capstone Design Project I

PerDrobe

Project Proposal

Personalized Wardrobe Outfit and Size Recommendation

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Academic Year 2026–2027

16 October 2026

1 Project Overview

PerDrobe is a proposed mobile platform for personalized wardrobe, outfit and clothing recommendations. It will combine a private digital wardrobe with explicit style and fit preferences and signals from likes, saves and follows. Users will be able to request outfits from clothes they own or find catalogue items that complement those clothes. Product-level size guidance will be provided only where the available measurements and product size information support it.

The engineering focus is a recommendation pipeline that represents garments, constructs candidates and ranks them for a particular user. Outfit Remix will adapt a discovered outfit to the user's wardrobe through compatible substitutions. We will compare a clothing-attribute baseline with configurations that add preference signals and then wardrobe context. The project spans CMPE 491 and CMPE 492; the application and experiments are planned work, and no measured results are reported in this proposal.

Project website: https://beratergun.github.io/perdrobe/
The academic project page hosts the proposal and will be updated throughout CMPE 491 and CMPE 492.

Problem and motivation

A clothing recommendation can reflect a person's taste while remaining difficult to use with the clothes that person owns. Size selection introduces a separate uncertainty: a familiar size label does not by itself establish how a particular product will fit. PerDrobe brings wardrobe context, style preferences and supported fit information into the same recommendation workflow.

Our intended users are people who want help assembling outfits from their own garments and evaluating possible additions to their wardrobe. Social discovery will provide outfit references and preference signals. It will support recommendation tasks without requiring the team to build a full social network.

Objectives and engineering contribution

We will build an integrated prototype that can maintain a wardrobe, rank owned-item outfits, adapt a reference outfit and recommend useful catalogue additions. A separate size module will provide evidence-based guidance on supported products. The system must preserve ownership boundaries and make missing information visible to the user.

The principal contribution will be a reproducible comparison of recommendation configurations under a common evaluation protocol. We will investigate how interaction-derived preferences and wardrobe context affect relevance, diversity and practical usefulness. The proposal makes no claim of being the first system to combine these ideas or of achieving an improvement before experiments are completed.

Research question

To what extent do style preferences derived from user interactions and personal wardrobe context improve clothing and outfit recommendation relevance over a clothing-attribute baseline?

2 Scope and Related Work

The core deliverable is a mobile prototype supported by a backend recommendation service and a bounded product catalogue. The following functions define the implementation scope across the two semesters.

Smart Wardrobe. Manual entry and editing of garments, with images and structured category, color, season and style attributes. Wardrobes will be private by default.

Personalized Outfit Recommendation. Rank compatible combinations from owned garments in Style What I Own mode, using explicit preferences and available interaction history.

Size and Fit Recommendation. Estimate a suitable size for supported catalogue products from optional measurements, usual sizes, fit preferences and available charts. Abstain when evidence is insufficient.

Outfit Remix. Map the garment roles in a discovered outfit to compatible items in the current user's wardrobe. Show substitutions and any unfilled role.

Wardrobe-Aware Shopping. In Complete My Wardrobe mode, rank catalogue items by personal relevance and the useful combinations they enable with owned garments.

Social Style Discovery. Share selected outfits, follow profiles or collections, like outfits and save collections. These actions will supply preference evidence; comments and messaging are outside the core scope.

Conditional extensions and exclusions

Duo Match is a conditional multi-user recommendation extension. It would seek two individually relevant outfits with compatible color, style or formality, using only information both users authorize. It may serve friends, siblings, partners or people attending the same event. It will be attempted only after the individual recommendation pipeline, core integration and evaluation preparation are complete.

Automatic clothing recognition, visual similarity search and automatic trend collection are optional investigations. Full virtual try-on, cloth simulation, advanced 3D avatar fitting, checkout, payment, comments and private messaging are outside the core scope. Manual garment attributes will remain sufficient for all core workflows.

Related work and positioning

Han et al. [1] study outfit compatibility and item completion using visual-semantic representations and a bidirectional recurrent model. This provides a technical reference for compatibility, but does not establish the benefit of personal wardrobe context in PerDrobe. He and McAuley [2] incorporate visual information into personalized ranking from implicit feedback; their work informs the possible use of representations alongside user signals.

Bayesian Personalized Ranking [3] provides a ranking objective for implicit feedback. It is a candidate learning approach if the project obtains sufficient interaction data. We will begin with a transparent attribute-based baseline and compare alternatives after inspecting data suitability. These references motivate design choices; no dataset or model has yet been selected for adoption.

3 Recommendation Methodology

Representation and candidate generation

Each garment will have a stable identifier, category, color, season and manually entered style attributes, with optional brand, size, fit and image fields. Missing attributes will be represented explicitly. User profiles will separate stated preferences from observations derived from interaction history. A new user will receive recommendations from onboarding choices and garment content before behavioral evidence is available.

For Style What I Own, the candidate generator will combine garments from the user's wardrobe into category-compatible outfits. It will limit the search by garment role and request constraints before scoring combinations. For catalogue discovery, it will retrieve items in relevant categories and retain source and availability metadata. The system will explain when no suitable candidate is found rather than silently substituting an unavailable item.

Compatibility and personalized ranking

The first implementation will use interpretable attribute similarity and compatibility rules. Configuration B will add agreement with stated preferences and signals from likes, saves, follows and remix actions. Interaction weights and any time decay will be selected on validation data. An absent interaction will not automatically be treated as dislike, and repeated actions will not be allowed to dominate the profile without normalization.

Configuration C will add features describing how a candidate relates to the user's owned garments. For shopping, these may include the compatible combinations an item enables and overlap with already owned items. For outfits, wardrobe context may capture garment reuse and diversity across recommendations. A weighted score is the initial design; a learned ranker or embedding representation will be considered only when the data supports a meaningful comparison. An LLM is not required for candidate generation or ranking.

Outfit Remix and wardrobe completion

Outfit Remix will represent a reference outfit as garment roles with attributes, retrieve possible substitutions for each role and rank the resulting combinations jointly. Independent nearest-item matches can produce an incoherent outfit, so the final score must also account for compatibility between substituted pieces. The output will identify reused items, substitutions and missing roles. In wardrobe-only mode a missing role will remain unfilled; catalogue suggestions will be offered separately in Complete My Wardrobe mode.

For example, a reference containing a white T-shirt, black jeans, white sneakers and an overshirt could be adapted using an owned white T-shirt, dark blue jeans and white sneakers. The overshirt would remain a gap. This is an illustrative workflow, not a generated result or a measured compatibility claim.

Size and fit estimation

The size module will normalize chart units and compare product-specific size ranges with voluntarily supplied measurements, usual sizes and fit preference. It will preserve distinctions between body measurements and garment dimensions. Where labels exist, a learned estimator can be compared with this chart-based baseline. Missing charts, ambiguous units or unsupported product categories will produce an insufficient-data response. A numerical confidence value will be used only if it can be calibrated against validation outcomes.

4 System Architecture and Data

The client will send authenticated requests to the backend. The backend will enforce permissions, obtain the permitted data and invoke the recommendation service. Models will be replaceable behind a stable service interface so experiments can compare ranking methods without redesigning the application.

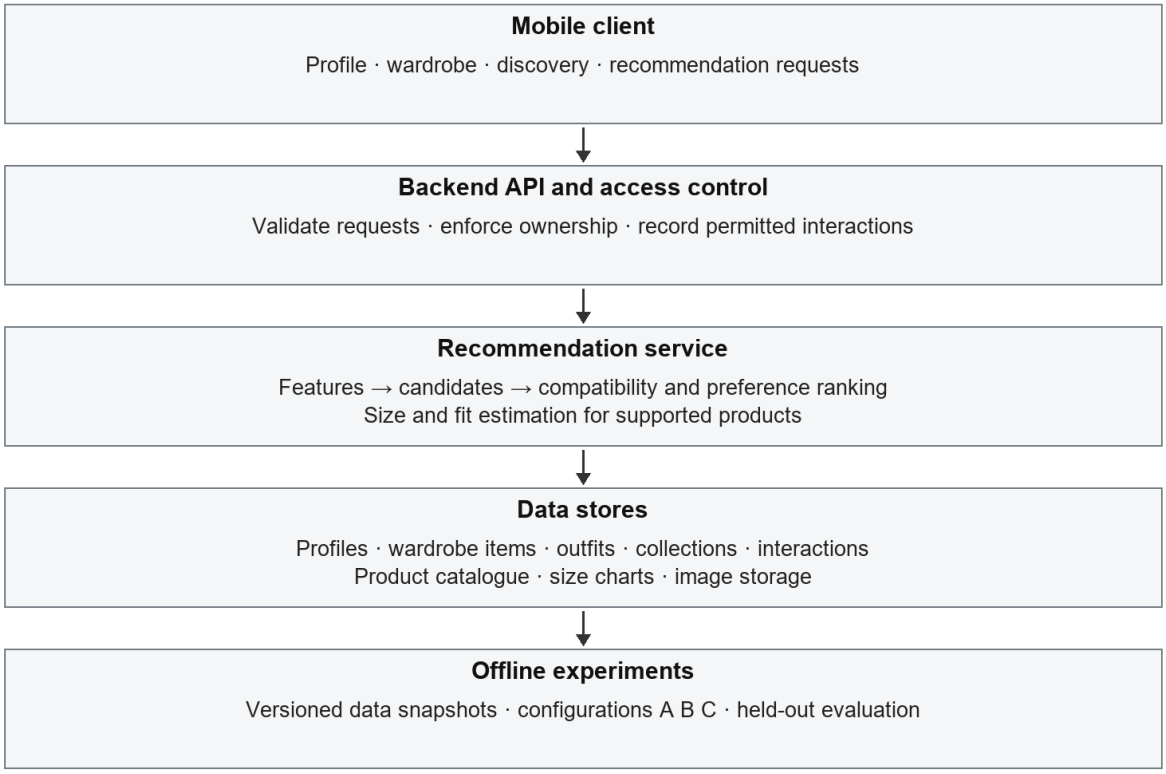


Figure 1. Proposed architecture. Arrows show the main processing sequence; the API mediates authorized reads and writes, and experiments use controlled data snapshots.

Data strategy and service boundaries

The relational data model will distinguish users, wardrobe items, catalogue products, outfits and their constituent items, collections, follows and timestamped interaction events. Ownership and visibility will be explicit. Product size charts will retain their source, unit and category. Image storage will be separate from structured records, with access governed by the associated item's visibility.

Initial development will use a manually prepared sample catalogue or a dataset whose license permits the intended use. No retailer API, partnership, dataset size or live stock feed is assumed. Data preparation will record provenance, permitted use, missing fields and transformations. Sample data will test software behavior; it will not be presented as evidence of real-user recommendation quality.

Proposed implementation direction

The initial direction is Android with Kotlin and Jetpack Compose, a Python FastAPI backend, PostgreSQL and separate object storage. Supabase services are one option for authentication and storage. Recommendation experiments may use scikit-learn or PyTorch; vector indexing will be added only if needed. Technology choices remain subject to prototyping. The academic website will use static HTML and CSS on GitHub Pages.

5 Experimental Design and Evaluation

To what extent do style preferences derived from user interactions and personal wardrobe context improve clothing and outfit recommendation relevance over a clothing-attribute baseline?

Model	Inputs	Purpose
A	Clothing attributes only	Reference ranking based on garment compatibility and content similarity.
B	A + explicit preferences and interaction signals	Estimate the incremental value of personalization.
C	B + personal wardrobe context	Estimate the incremental value of wardrobe compatibility and reuse.

A, B and C will use the same representation, request constraints and eligible candidate pool within each task. Wardrobe-only candidates come from owned items in every configuration; C adds wardrobe features to ranking. Catalogue comparisons will use the same snapshot, avoiding gains caused by different candidate eligibility.

Protocol and relevance judgments

We will evaluate outfit ranking, catalogue recommendation and remix adaptation separately. With sufficient timestamped interactions, earlier events will support training, a later interval tuning and a final interval held-out testing. Features will use only information available before each cutoff. Duplicate outfits and related item variants will be checked across splits to reduce leakage.

A consent-based user evaluation is planned, subject to supervisor agreement and applicable university processes. Recruitment and participant count are not yet fixed. Participants will assess shuffled recommendations without configuration labels. A pilot will establish wearable/useful relevance criteria and a graded scale, frozen before final testing. Explicit judgments will be distinguished from implicit signals; missing feedback will not count as dislike.

Area	Planned measures and interpretation
Ranking relevance	Precision@K for a predefined binary relevance threshold; NDCG@K for graded judgments. K and the rating rubric will be fixed before final testing.
Practical usefulness	User ratings of outfit wearability, remix faithfulness and usefulness of catalogue additions. Report these tasks separately.
Diversity and coverage	Attribute-based intra-list diversity and the fraction of eligible catalogue items recommended across evaluated requests.
Size and fit	Exact-size or fit-category agreement on labeled, supported cases, alongside the supported-case and abstention rates. Charts alone are not ground-truth fit labels.
System performance	Median and 95th-percentile API latency under a documented workload, separating inference from network and storage time.

Analysis and limits

We will report per-user differences and uncertainty where the sample permits it, separating cold-start cases. A further ablation may separate onboarding from interaction-derived preferences. Conclusions will be limited by the available judgments and fit labels; no configuration is assumed to outperform another.

6 Development Plan and Privacy

We propose short implementation iterations with a shared issue backlog, version control and reviewed integration changes. Work will cover mobile development, backend/data services and recommendation experiments, with cross-review between these areas. Individual roles are not yet assigned. Each iteration will produce a working increment and record design decisions and remaining risks.

Course	Phase	Planned work	Completion evidence
CMPE 491	Early phase	Refine requirements and related work; select permitted data; define schemas and evaluation protocol.	Reviewed scope, data inventory and testable requirements.
CMPE 491	Middle phase	Build profile, wardrobe, catalogue and interaction services; integrate the mobile foundation.	Integrated vertical slice with access controls.
CMPE 491	Final phase	Implement configuration A and initial size-chart logic; verify logging and initial recommendation flow.	Baseline prototype, design documentation and evaluation plan.
CMPE 492	Early phase	Implement B and C; integrate Outfit Remix and wardrobe-aware shopping.	Comparable ranking configurations and core workflows.
CMPE 492	Middle phase	Refine supported size guidance; run pilot evaluation; consider Duo Match only after core integration.	Frozen experiment protocol and tested core prototype.
CMPE 492	Final phase	Run comparisons and user evaluation; test performance and privacy; stabilize the demonstration.	Final report, presentation, reproducibility materials and updated website.

These phases are indicative, not official course deadlines. Detailed dates will follow the course calendar and supervisor feedback. The CMPE 491 baseline, reliable data pipeline and access controls are prerequisites for the CMPE 492 comparison. Optional extensions will not displace final evaluation or core integration.

Privacy and security design

Wardrobes will be private by default. Sharing an outfit will require an explicit user action and will not publish body measurements or the rest of the wardrobe. The prototype will minimize collected information and let users edit or delete their profile, measurements, garments and images. Duo Match, if implemented, will use only mutually authorized items and preferences.

The backend will validate authentication and resource ownership on every protected operation. Planned controls include encrypted transport, protected image access, file-type and upload-size validation, and exclusion of measurements and credentials from application logs. Tests will include attempts to read or modify another user’s records. Experiment logs will use pseudonymous identifiers and a defined retention period agreed before participant data collection.

Fit guidance will be expressed without body-value judgments. Users will be able to correct inferred preferences, and the evaluation will examine whether recommendations repeatedly favor narrow styles or popular items. We will document permissions and university requirements before collecting participant data; this proposal does not assert legal or institutional approval.

7 Risks and Expected Deliverables

Risk	Mitigation
Data access and licensing	Use a bounded, manually prepared or permitted catalogue; record provenance and reuse conditions. Retailer integration is not a dependency.
Sparse preference history	Use onboarding and content similarity for new users; report cold-start and returning-user results separately.
Incomplete size information	Normalize units, retain product/category context and abstain when charts or labels are inadequate.
Scope and integration	Protect the recommendation comparison and core workflows; defer Duo Match and visual extensions. Integrate early through versioned API contracts.
Biased or weak evaluation	Use matched candidate pools, held-out judgments and per-user summaries; report limitations and negative findings.
Private data exposure	Test ownership checks, image access, deletion and consent boundaries before participant use.

Deliverables and completion criteria

By the end of CMPE 491, we aim to deliver the requirements and design documentation, data inventory, integrated wardrobe/profile foundation, initial interaction storage, content baseline and evaluation protocol. CMPE 492 will deliver the integrated core prototype, recommendation comparisons, supported size module, final report and demonstration. The project website will retain the proposal and add later reports and presentations at the same address.

Core acceptance will require a user to add and manage garments, obtain an owned-item outfit, remix a reference outfit with explicit substitutions, and inspect a catalogue addition in wardrobe context. Supported size requests must show their basis; unsupported requests must abstain. Experiment materials will record the data version, split, configuration, random seed where applicable and metric definitions. Privacy tests and reproducible evaluation scripts will accompany the final implementation.

The expected outcome is evidence about the usefulness and limitations of adding preference signals and wardrobe context, together with an integrated prototype that demonstrates those design choices. Improvements in relevance, fit accuracy or wardrobe use are hypotheses to assess, not promised results. Any incomplete core task or deferred extension will be reported explicitly in the final documentation.

References

[1] X. Han, Z. Wu, Y.-G. Jiang and L. S. Davis. Learning Fashion Compatibility with Bidirectional LSTMs. ACM Multimedia, 2017.
https://arxiv.org/abs/1707.05691

[2] R. He and J. McAuley. VBPR: Visual Bayesian Personalized Ranking from Implicit Feedback. AAAI, 2016.
https://arxiv.org/abs/1510.01784

[3] S. Rendle, C. Freudenthaler, Z. Gantner and L. Schmidt-Thieme. BPR: Bayesian Personalized Ranking from Implicit Feedback. UAI, 2009.
https://arxiv.org/abs/1205.2618